

## Summary

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Multidisciplinary Mechanical Engineer (EIT) with 3 years of process engineering experience in an H<sub>2</sub>O<sub>2</sub> production plant, specializing in DCS operations, vacuum distillation, and SOP-driven chemical process control. Additionally brings 2+ years of automotive manufacturing experience in HVAC subassembly, vehicle seat production, chassis design, and EPC project execution. Proficient in Revit, AutoCAD, and SolidWorks for system modeling and design analysis. Lean Manufacturing and Six Sigma Green Belt certified, committed to delivering efficient, reliable, and sustainable engineering solutions.

## Education

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| <b>Ph.D. in Engineering</b>   Louisiana Tech University   CGPA: 3.85/4.00                                 | May 2026     |
| <b>M.Sc. in Mechanical Engineering</b>   Louisiana Tech University   CGPA: 3.92/4.00                      | August, 2023 |
| <b>B.Tech. in Mechanical Engineering</b>   National Institute of Technology, Jamshedpur   CGPA: 3.49/4.00 | June, 2016   |

## Professional Certificates & License

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| • Engineer Intern, EI 0036277         | • Lean Manufacturing                             |
| • Passed FE in Mechanical Engineering | • Project Management Professional (PMP) Training |
| • Six Sigma Green Belt                | • PE Mechanical (HVAC & Refrigeration) Passed    |

## Technical Skills

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**Material Characterization:** SEM w/ EDS, LSCM, Zeta potential analyzer, AFM, TGA.

**Process Engineering and Optimization:** Process design and control, Root cause analysis (RCA), Process standardization, DMAIC, 5S, Fishbone diagram, Pareto diagrams, Control charts, FMEA, Process mapping, Statistical Process Controls (SPC), Design of Experiments (DOE).

**Continuous improvement:** Lean manufacturing, Kaizen, Poka-yoke, VSM, Kanban, DFM, Product development techniques, Visual management, KPI tracking, Cross-functional project leadership, Capacity planning.

**Computational tools and programming:** MATLAB, MS Excel, LAMMPS, Polymatic, VMD, SolidWorks, AutoCAD, REVIT, SQL, Power BI, Minitab, SAP S/4HANA, Zoho.

**Code and Standards:** Pressure (ASME), HVAC (ASHRAE), Quality (ISO), Materials (ASTM), Automotive (SAE).

## Professional Experience

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| <b>Graduate Researcher, Louisiana Tech University</b>   Ruston, LA. | 2021 – 2026 |
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- Performed Conducted over 2000+ molecular dynamics simulations (atomistic and coarse-grained) to evaluate asphalt-polymer miscibility, analyze phase behavior (spinodal and binodal), and model oxidative aging effects based on experimentally informed chemistry.
- Disseminated key research findings through oral and poster presentations at multiple graduate symposiums and national conferences. Published articles in peer-reviewed journals.

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| <b>Process Engineer, Hydrogen peroxide plant</b>   Al-Razi Chemical Complex Limited | 2018 – 2020 |
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- Monitored and optimized continuous chemical process using DCS in Vacuum Distillation and Solvent Recovery units.
- Executed start-up and shutdown of chemical process through the DCS in accordance with approved SOPs,
- Supervised operators and ensured compliance with SOPs, HSE guidelines, and safe chemical handling practices.
- Monitored utility systems including boilers, DM water plant, cooling towers, and auxiliary systems.
- Prepared daily production and shift reports, tracked KPI, and communicated operational recommendations to senior engineers and plant managers.
- Participated in annual shutdown planning, equipment inspection, maintenance coordination, and post shutdown start-up activities.

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| <b>Assistant Mechanical Engineer, Quality Control</b>   AFTAB Automobiles LTD. | 2016 – 2017 |
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- Performed HVAC sub-assembly quality validation, including leak testing, pressure integrity checks, and airflow verification, consistent with GMP requirements.
- Supervised quality validation for 3,000+ bus seat assemblies, ensuring structural integrity and occupant safety in accordance with automotive manufacturing and regulatory standards.
- Inspect fabrication materials prior to start-up, changeovers or when handing off to next shift of fabrication process.
- Complete periodic line checks to determine compliance with interior and exterior structures' specification.
- Evaluate discarded structures and recommend disposal measures.